Dr. Zoe Molitor (she, they)

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Education

PhD **MIT — Earth, Atmospheric and Planetary Sciences (EAPS)**, Geology
Cambridge, MA
Jan 2018 – Jan 2024

BS **Purdue University**, Geology and Geophysics
West Lafayette, IN
Jan 2015 – Jan 2018

Experience

Boston College — Earth and Environmental Sciences, Postdoctoral Researcher
Chestnut Hill, MA
Jan 2026 – present
8 months

Yale University — Earth and Planetary Sciences (EPS), NSF EAR Postdoctoral Fellow
New Haven, CT
Jan 2024 – Jan 2026
2 years 1 month

Boston College, Adjunct Professor — Structural Geology, EESC 3385
Fall 2026 semester.
Chestnut Hill, MA
Jan 2026 – Jan 2026
1 month

MIT, Graduate Teaching Assistant, 12.002
Spring 2021 and Spring 2022 semesters.
Cambridge, MA
Jan 2021 – Jan 2022
1 year 1 month

Purdue University, Undergraduate Teaching Assistant, EAPS 12000
West Lafayette, IN
Jan 2017 – Jan 2017
1 month

Volunteer

LGBTQ+ Working Group, Founder and Lead Organizer
Yale University
Jan 2024 – Jan 2026

WEPSY (Women and Nonbinary People in EPS), Organizing Committee Member
Yale University
Jan 2024 – Jan 2026

EPS Department IDEA Committee, Member
Yale University
Jan 2024 – Jan 2026

SURF Program, Research Mentor — Brynn Graham (Undergraduate, Mt. Holyoke College)
Summer 2025 – Spring 2026.
Yale University
Jan 2025 – Jan 2026

Pathways Summer Scholars, Lead Organizer, Earth Science Week-Long Workshop
Jan 2026 – present

Pathways Summer Scholars High School Program, Co-Organizer, Earth Science Week-Long Workshop
Yale University
Jan 2025 – present

West Campus Science Festival, Group Leader, Chemistry Rotation
Yale University
Jan 2024 – present

LINK12 K–12 Outreach Program, Administrative Member — EAPS, MIT, Jan 2022 – Jan 2024

Letters to a PreScientist, Pen Pal (K–12 Program) — Jan 2022 – present

DEAPS Freshman Orientation, Field Trip Leader — Geology of the White Mountains — Jan 2022 – present

Boston826, Hydraulic Fracturing Demonstration & Interview with Local K–12 Students — Jan 2021 – present

DEAPS Freshman Orientation, Field Trip Leader — Geology of Yellowstone — Jan 2020 – present

Publications

Extreme Moho Relief Preserved in Ohio's Crust — Jan 2026
Z. Molitor, F. Link, M. Long, J. Ague
(GSA Today, 36, 4–9)

A whole-rock geochemistry-based semi-quantitative proxy for magmatic oxygen fugacity — Jan 2026
Z. Liu, O. Jagoutz, H. Rezeau, H. L. L. Zhang, B. Klein, Z. Molitor, et al., D. C. Zhu
(Earth and Planetary Science Letters, 673, 119742)

Bedrock Geology of NE Connecticut: A Transect from the Bronson Hill Anticlinorium to Putnam-Nashoba Terrane — Jan 2025
Z. Molitor, J. Ague, B. Graham
digitalmaine.com/geo_docs/204 (In A.D. Rooney & M.D. Long (Eds.), Guidebook for Field Trips in Southwestern Connecticut (Trip D, pp. 86–112), Geology Documents)

Permo-Carboniferous crustal structure and state of strain in the New England Appalachians, USA — Jan 2025
Z. Molitor, O. Jagoutz, A. Moser, A. Cruz-Uribe, A. Cross
(Geosphere, 21(6), 991–1019)

Crustal section in the New England Appalachians records polybaric fractional crystallization-driven magma differentiation — Jan 2025
Z. Liu, O. Jagoutz, Z. Molitor, J. Ramezani, D. C. Zhu
(Earth and Planetary Science Letters, 671, 119647)

The viscosity of a partially molten layer in a paleo-orogenic plateau — Jan 2024
Z. Molitor, T. Mittal, O. Jagoutz
(Earth and Planetary Science Letters, 648, 119060)

ReversePetrogen: A Multiphase Dry Reverse Fractional Crystallization-Mantle Melting Thermobarometer Applied to 13,589 Mid-Ocean Ridge Basalt Glasses — Jan 2021
S. Brown Krein, Z. Molitor, T. L. Grove
(Journal of Geophysical Research: Solid Earth, 126(8), e2020JB021292)

Testing models of orogenic conjugate shear in the New England Appalachians
Z. Molitor, O. Jagoutz, A. Cross, A. Moser
(In preparation)

Invited Talk: Extreme Moho Relief Beneath the State of Ohio — Nov 2025
Invited talk.
Z. Molitor
(Lahmont Doherty Earth Observatory, Geodynamics Seminar)

Spatio-Temporal Controls on Orogenic Thermal Structure and Evolution in the Southern New England Appalachians Mar 2026
Conference abstract, 22–23 Mar 2026.
Z. Molitor, J. Ague, B. Graham, J. Eckert, A. Cruz-Uribe
(Abstract 29-5, 2026 Northeast GSA)

Assessing the Impact of High Temperature Metamorphism and Partial Melting on the Viscosity of the Tatnic Hill Formation Oct 2025
Conference abstract, 19–22 Oct 2025.
B. Graham, Z. Molitor, J. Ague
(Abstract 284-7, 2025 GSA Connects)

Spatio-Temporal Controls on Orogenic Thermal Structure and Evolution in the Southern New England Appalachians Oct 2025
Conference abstract, 19–22 Oct 2025.
Z. Molitor, J. Ague, B. Graham, J. Eckert
(Abstract 284-7, 2025 GSA Connects)

Constraining the 3D Geometry of Conjugate Orogenic Strike-Slip Shear Zones in the New England Appalachians Mar 2024
Conference abstract, 17–19 Mar 2024.
Z. Molitor, O. Jagoutz, A. Cross, A. Cruz-Uribe, A. Moser
(Abstract 20-6, 2024 Northeast GSA)

Testing Models of Conjugate Faulting in the Late Paleozoic New England Appalachian Orogen Dec 2022
Conference abstract, 12–16 Dec 2022.
Z. Molitor, L. Royden, O. Jagoutz
(Abstract T55B-0068, 2022 AGU Fall Meeting)

A New Model for Dynamic and Thermal Compensation around Mantle Plumes: Implications for Volume and Heat Flux Dec 2021
Conference abstract, 13–17 Dec 2021.
Z. Molitor, L. Royden, O. Jagoutz
(Abstract DI25B-0027, 2021 AGU Fall Meeting)

Mantle Plume – Spreading Ridge Interaction: A Comparative Study of the Red Sea and Reykjanes Ridges May 2020
Conference abstract, 4–8 May 2020.
Z. Molitor, O. Jagoutz, L. Royden, S. Brown, G. Port, A. Dogru, J. Keller
(Abstract EGU2020-11138, 2020 EGU General Assembly)

Field Experience

Spring 2026: Nahant and Salem, MA — 3 Days

Fall 2025: NEIGC, Connecticut — Field Trip Leader

Fall 2024: Connecticut, Massachusetts, and New Hampshire — 6 Weeks

Jun 2023: Sicily, Italy (MIT Department Field Trip) — 1 Week

Summer 2022: New England Appalachians (NH, MA, CT) — 4 Weeks

Jan 2022: Mojave Desert (MIT Field Camp) — 2 Weeks

Dec 2021: Vermont and New Hampshire — 1 Week

Nov 2021: Newfoundland, Canada — 2 Weeks

Jun 2021: New Hampshire — 1 Week

Fall 2020: New Hampshire — 4 Weeks

Jan 2019: Mojave Desert (MIT Field Camp) — 2 Weeks

Jun 2018: Sardinia, Italy (Undergraduate Field Camp) — 4 Weeks

Mar 2018: Utah (Undergraduate Field Methods) — 1 Week

Technical Skills

Programming Languages: Python, MATLAB (MTEX)

Computer Programs: ArcGIS, Adobe Illustrator, Inkscape, Perple_X, TheriakDomino, Microsoft Office Suite

Laboratory Analysis: SEM/EPMA (Electron Probe Microanalyzer), SEM/EBSD (Electron Backscatter Diffraction), Heavy Mineral Separation, LA-ICP-MS